

Diagnostic

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This is also available at the following remote repository: <https://github.com/m-mironova/ps446>

By far the biggest issue in using a balance table for this paper is the number of categorical variables in it: sex, race, state, etc. Though factorization is possible, it was not executed here, but would have enabled R to more accurately represent the data and measure all covariates, not just numerical data. Race would still have presented an issue, however, since for many individuals, there are multiple values listed. Potentially, this would still have been possible to adjust for. As it stands, the balance table reports the numerical demographic information on the experiment participants only.

The code for the following table is accessible under the diagnostic.R file.

Balance Measures

	Type	Diff.Un	V.Ratio.Un	KS.Un
yob	Contin.	-0.0110	1.0380	0.0322
educ	Contin.	-0.0439	1.0105	0.0333
ideo	Contin.	-0.0118	1.0769	0.0325
ideo:<NA>	Binary	0.0006	.	0.0006
policy_educ	Contin.	0.0405	1.1162	0.0134
policy_tax_V1	Contin.	0.0016	0.9868	0.0094
policy_tax_V1:<NA>	Binary	0.0293	.	0.0293
policy_tax_V2	Contin.	0.0167	1.0544	0.0115
policy_tax_V2:<NA>	Binary	-0.0293	.	0.0293
policy_immi	Contin.	0.0033	0.9858	0.0085
policy_mari	Contin.	0.0315	1.0269	0.0135

Sample sizes

	Control	Treated
All	1060	3174

I am unsure of where the covariate values appeared from, as they were not specified in the code used to create the balance table. As I am not clear on the meaning of these lines, they are ignored for the rest of the assignment.

The variables reported in the balance tables are as follows:

Table 1: Balance Table Variables

Variable Name	Meaning (from codebook)
yob	Respondent's age (year 2006 = 1; year 1920 or before = 87).
educ	Respondent's highest level of school completed.
ideo	Respondent's self-reported ideology (more conservative = higher).
policy_educ	Respondent's policy position for education finance (more conservative = higher).
policy_tax_V1, policy_tax_V2	Respondent's policy position for tax rate (more conservative = higher). [V2: for respondents residing in a state without a state income tax].
policy_immi	Respondent's policy position for immigration (more conservative = higher).
policy_mari	Respondent's policy position for marijuana (more conservative = higher).

The columns represent: Diff.Un = standardized mean difference, V.Ratio.Un = variance ratio, and KS.Un = Kolmogorov-Smirnov statistic. Note that all of these are unadjusted (.un), meaning that they use the raw data provided from the study's supplemental materials, with no matching, regression, accounting for confounders, or other manipulations performed on them.

Note that all standardized mean difference values (the difference in the mean between the control and treated groups) are within $|x|=0.04$, meaning that the randomization is relatively well balanced. That is, no treatment group is further than 0.04 standard deviations away from the control group on any of the specified covariates.

The variance ratio similarly is within the acceptable norm, where x belongs to $[0.9858, 1.1162]$. Given that 1 is perfect balance (where 1 means the treatment explains 100% of the variation between the treated and controlled groups), this is similarly good news for the study.

Finally, the Kolmogorov-Smirnov statistic is also in good order, the biggest difference between the treated and untreated distribution, by covariate, is 0.0333 and the smallest is 0.0085. This also satisfies all the necessary conditions to trust the randomization of the study.

In sum, the balance tables report a level of confidence similar to the equivalence ratio table reported in the study.

Links to major sources relied upon for this assignment:

<https://www.stat.berkeley.edu/~s133/factors.html>

<https://cran.r-project.org/web/packages/MatchIt/vignettes/assessing-balance.html>

https://lost-stats.github.io/Presentation/Tables/Balance_Tables.html

<https://www.rdocumentation.org/packages/matchMulti/versions/1.1.12.1/topics/balanceTable>

<https://cran.r-project.org/web/packages/cobalt/vignettes/cobalt.html> <https://ngreifer.r-universe.dev/cobalt>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11562970/>

<https://www.sciencedirect.com/topics/computer-science/variance-ratio>